

ACESSIBLE AREA + VARIABLE SELECTION

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University of Florida

FLORIDA MUSEUM™

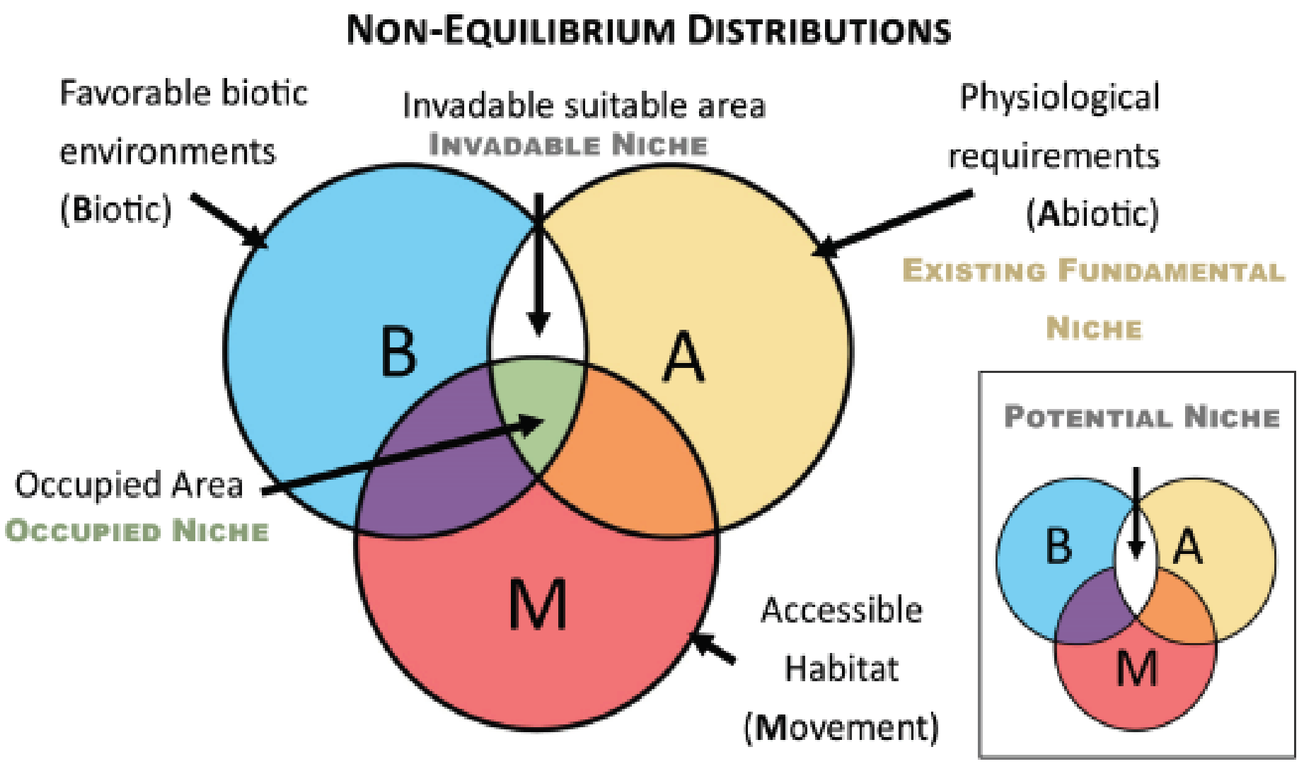
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1.Defining Accessible Area

- Accessible area can be defined based on:
  - species ecology
  - dispersal ability
  - geographical barriers
- **Accessible area (M)** defines where a species *could* have dispersed to, given geography and history.



Brown and Carnaval. 2019. A tale of two niche: methods, concepts, and evolution. Frontiers of Biogeography.

1.The Role of the Accessible Area (M) in ENM

The crucial role of the accessible area in ecological niche modeling and species distribution modeling

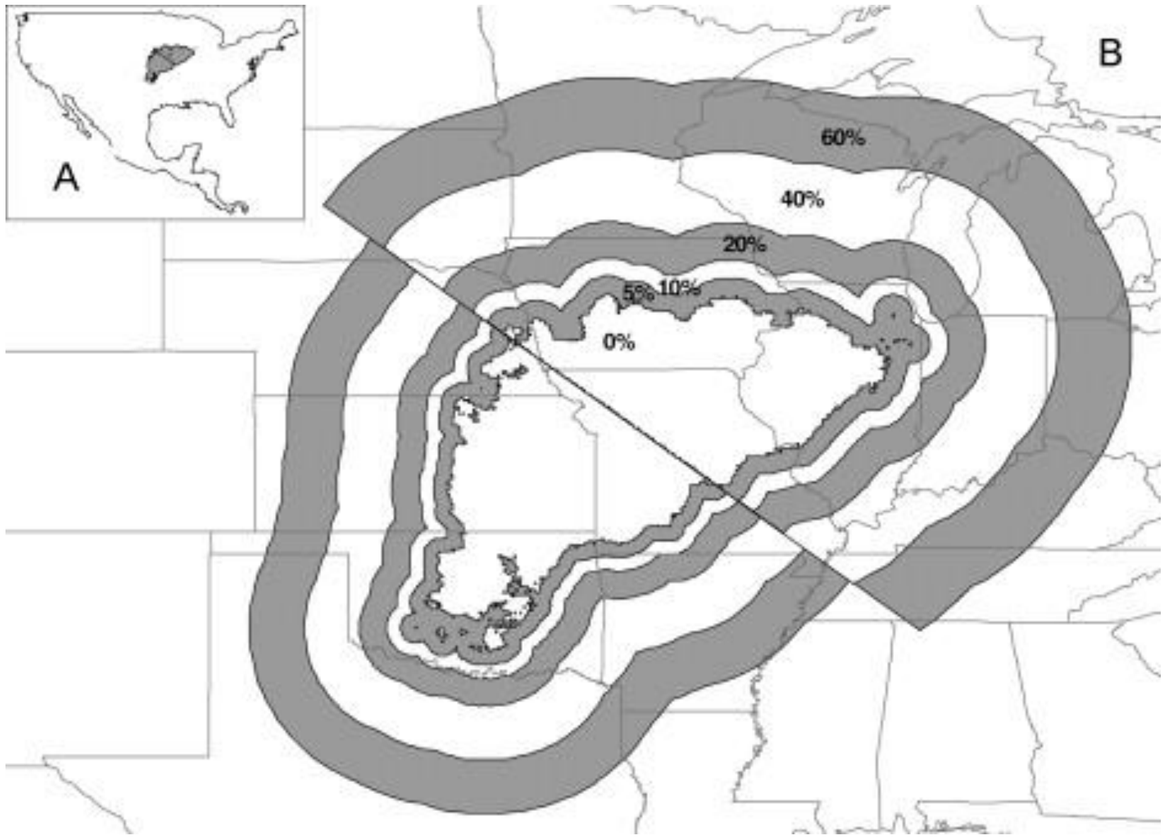
Narayani Barve [a], Vijay Barve [a, 1], Alberto Jiménez-Valverde [a, 1], Andrés Lira-Noriega [a], Sean P. Maher [a, 2], A. Townsend Peterson [a], Jorge Soberón [a], Fabricio Villalobos [b]

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- Add a buffer around the species distribution to capture potential movement.

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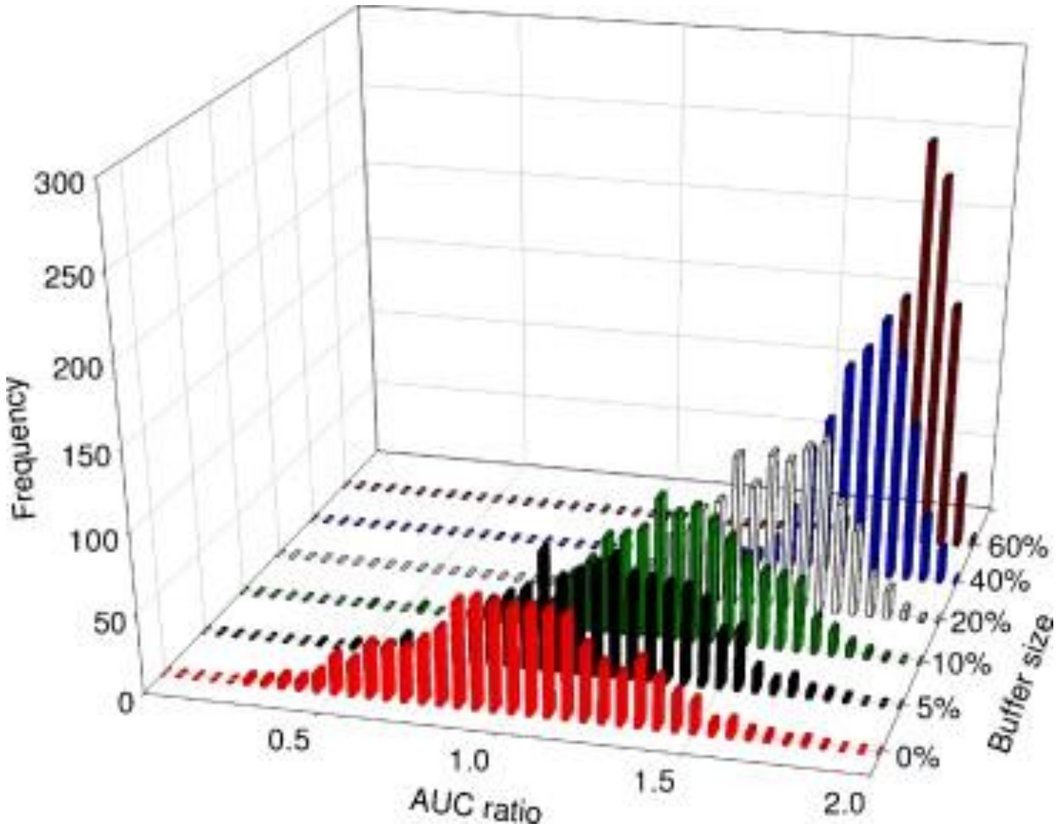
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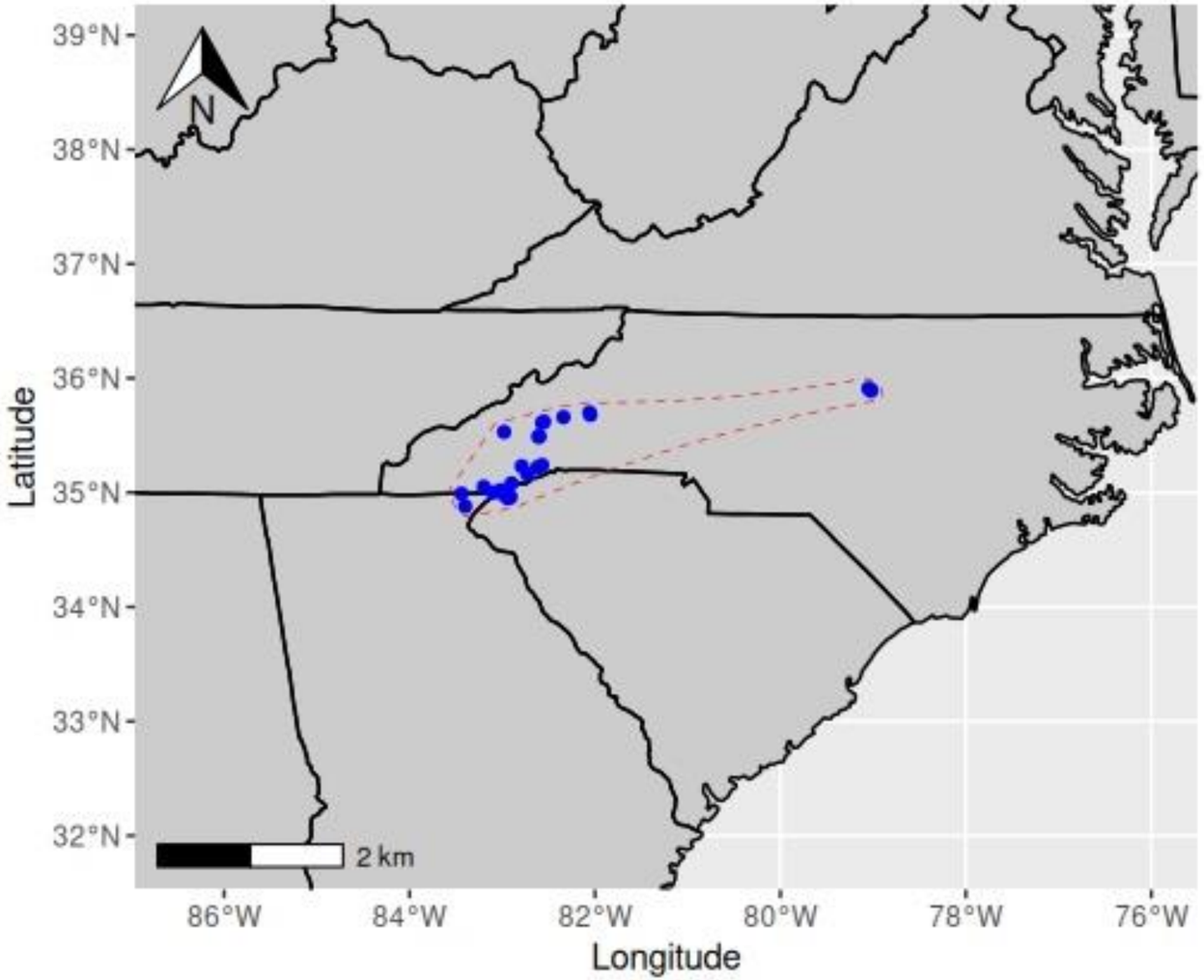
- More area, AUC increased.
- Higher model significance as buffer size increased.

1.Defining the Accessible Area in Practice

1. **Create spatial boundary around species occurrences.**
2. Create a Spatial Buffer around occurrence boundary.

   → Measure pairwise distances between species occurrences.

   → Extract the 80th percentile distance as a buffer radius.
3. Obtain gridded abiotic data layers.
4. Crop layers to the shared accessible area.
5. Select layers for ecological niche modeling.



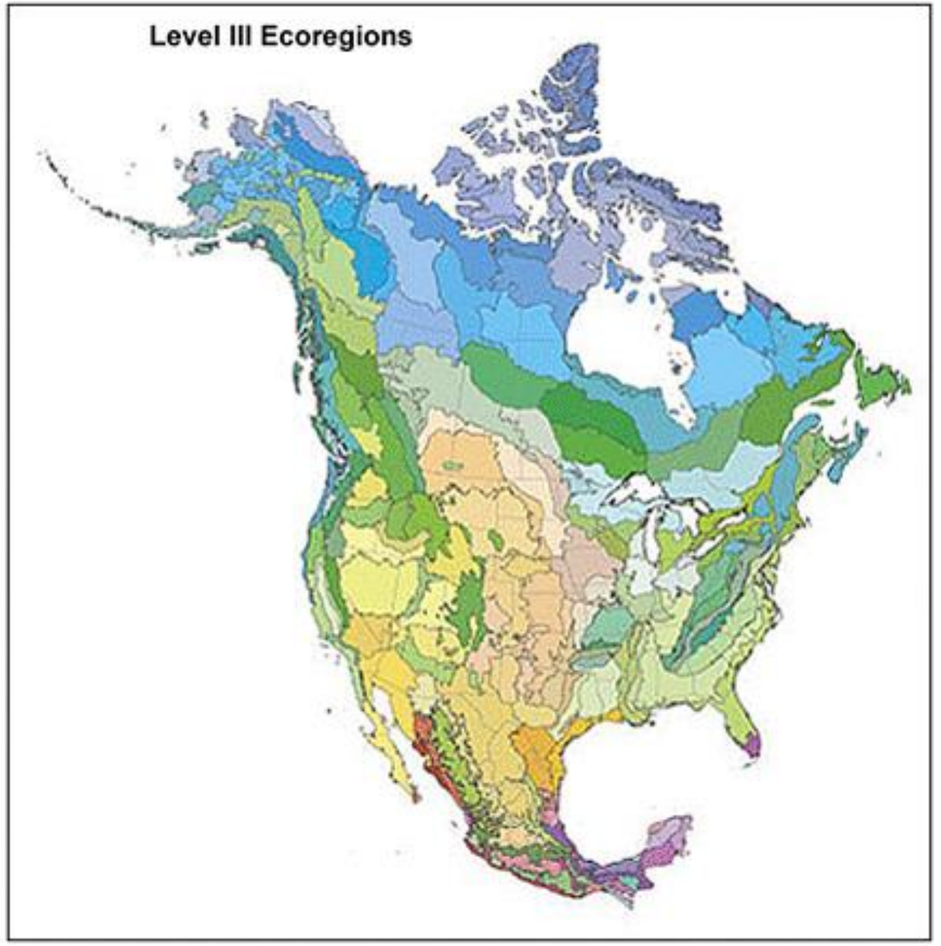
1.Multiple ways to create species boundaries

- **Convex Hull** - Encloses all points in the smallest convex polygon.
    - Equivalent to the boundary formed by stretching a rubber band around the outermost points.

- **Alpha Hull** - A flexible, concave boundary estimator controlled by an alpha value.
    - Captures non-convex patterns and gaps in occurrence data (e.g., disjunct populations, island distributions). <- we do this today

- **Ecoregions** - Identify clear distributional boundaries and ecoregions using the World Wildlife Fund Terrestrial Ecoregions (see Rautsaw et al. 2022).

- **Point-based** - Defines species' accessible area directly from occurrence coordinates.
    - Constructs boundaries (e.g., convex or alpha hulls) and buffers using the spatial structure of the occurrence points (see Melton et al. 2022).

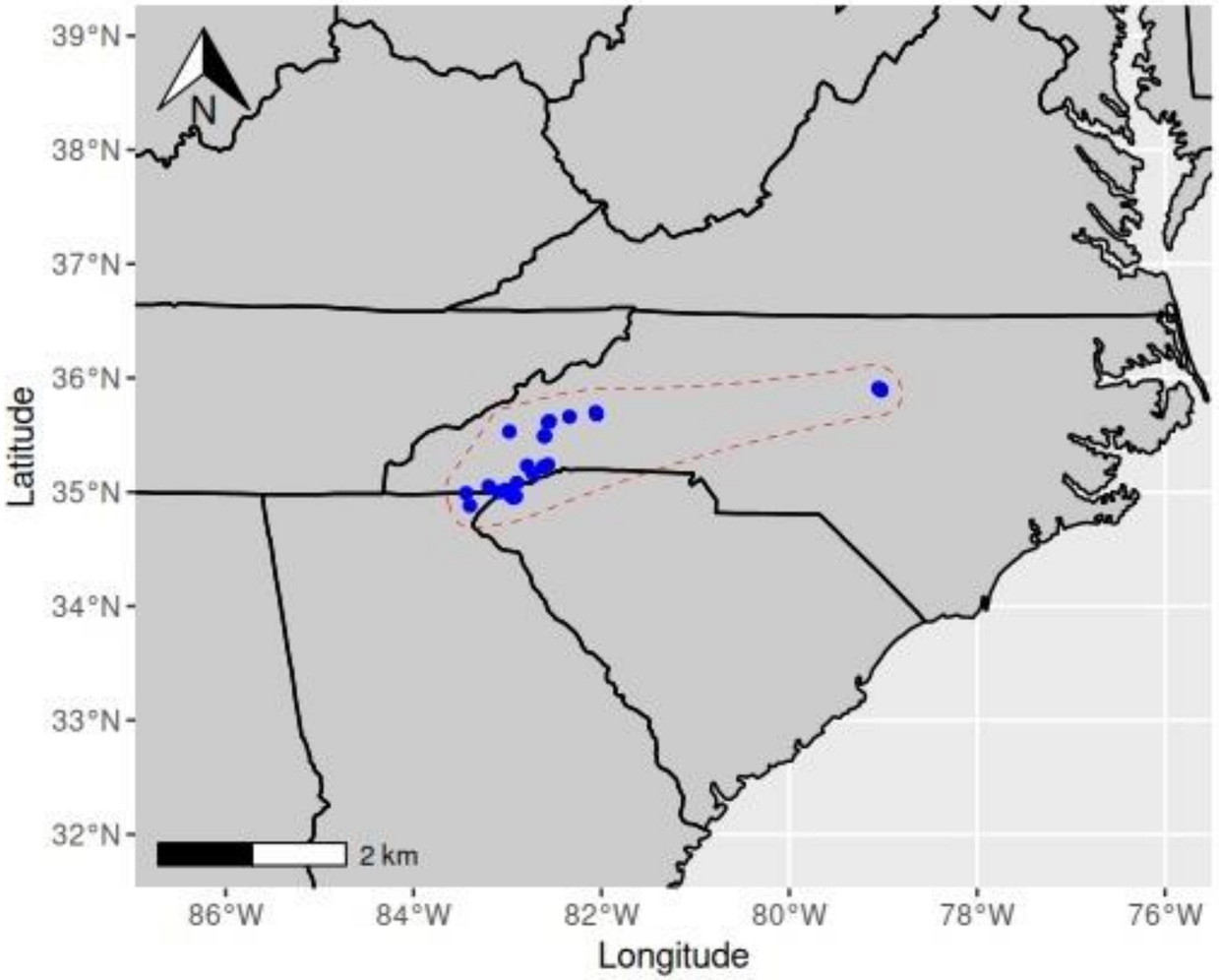


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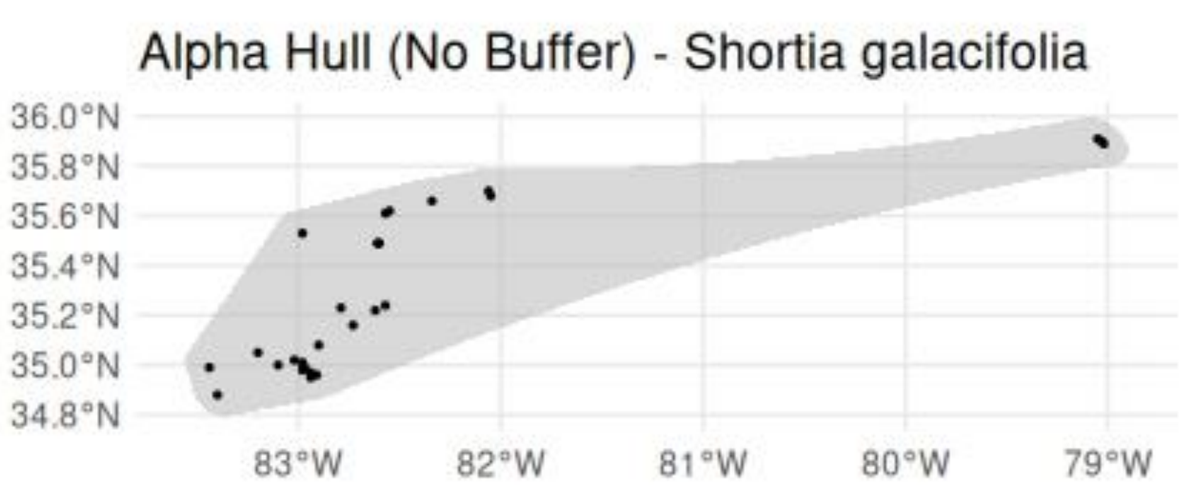
1. Create spatial boundary around species occurrences.
2. **Creating a Spatial Buffer Around Occurrence Alpha Hull**

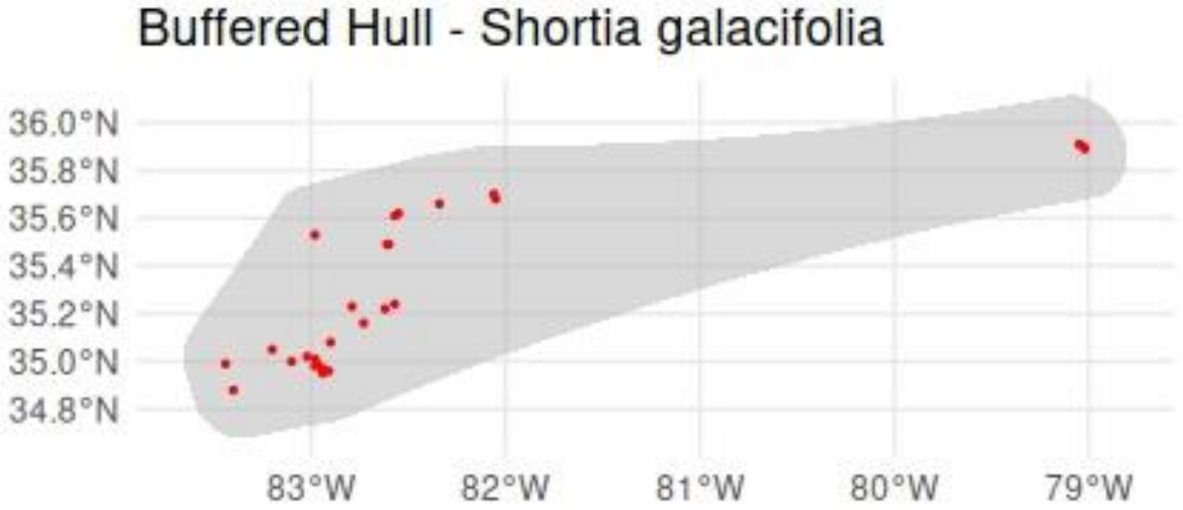
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   → Extract the 80th percentile distance as a buffer radius.
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5. Select layers for ecological niche modeling.



1.Defining the Accessible Area in Practice





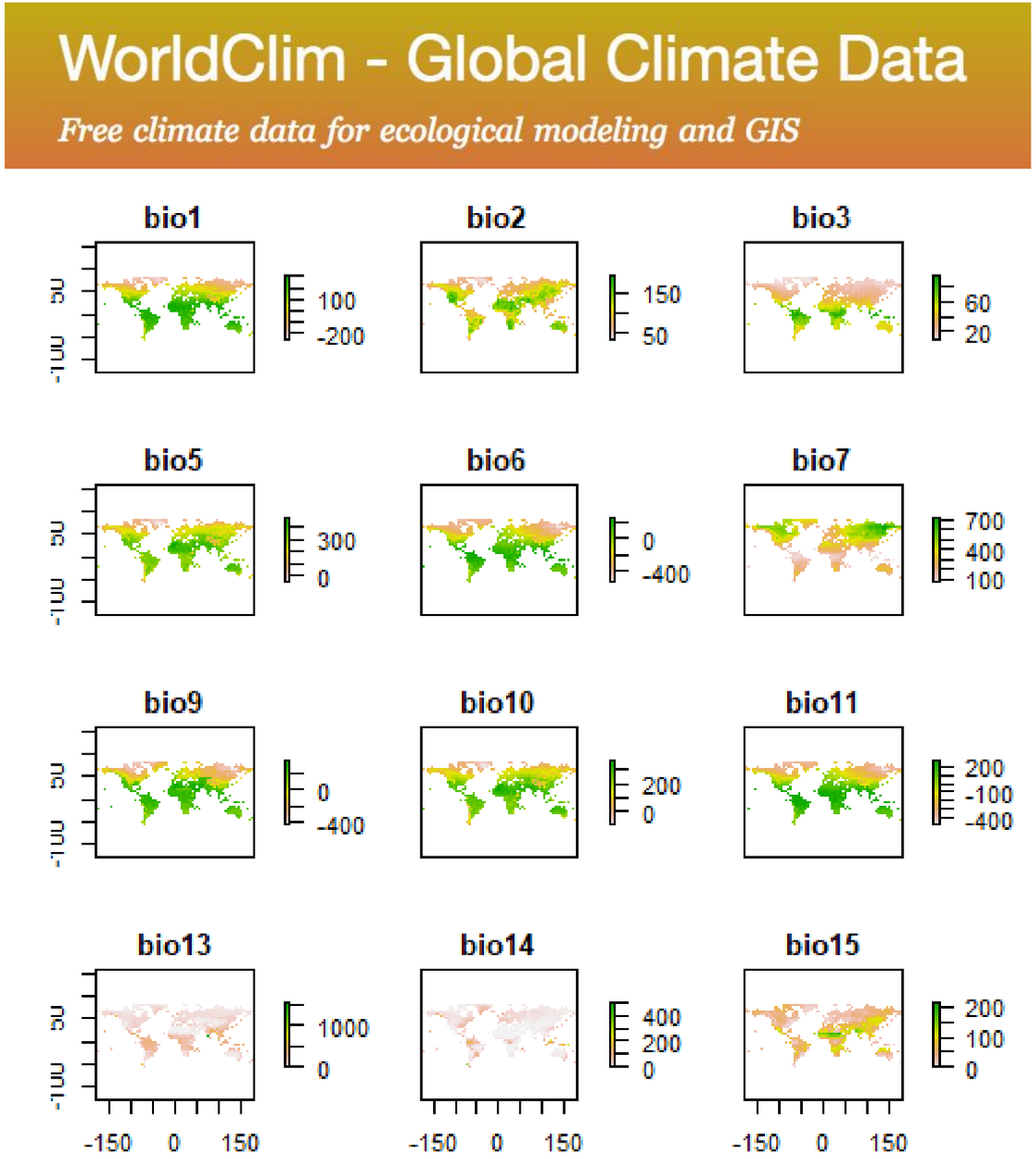
1.Defining the Accessible Area in Practice

1. Creating a Spatial Buffer Around Occurrence Convex Hull

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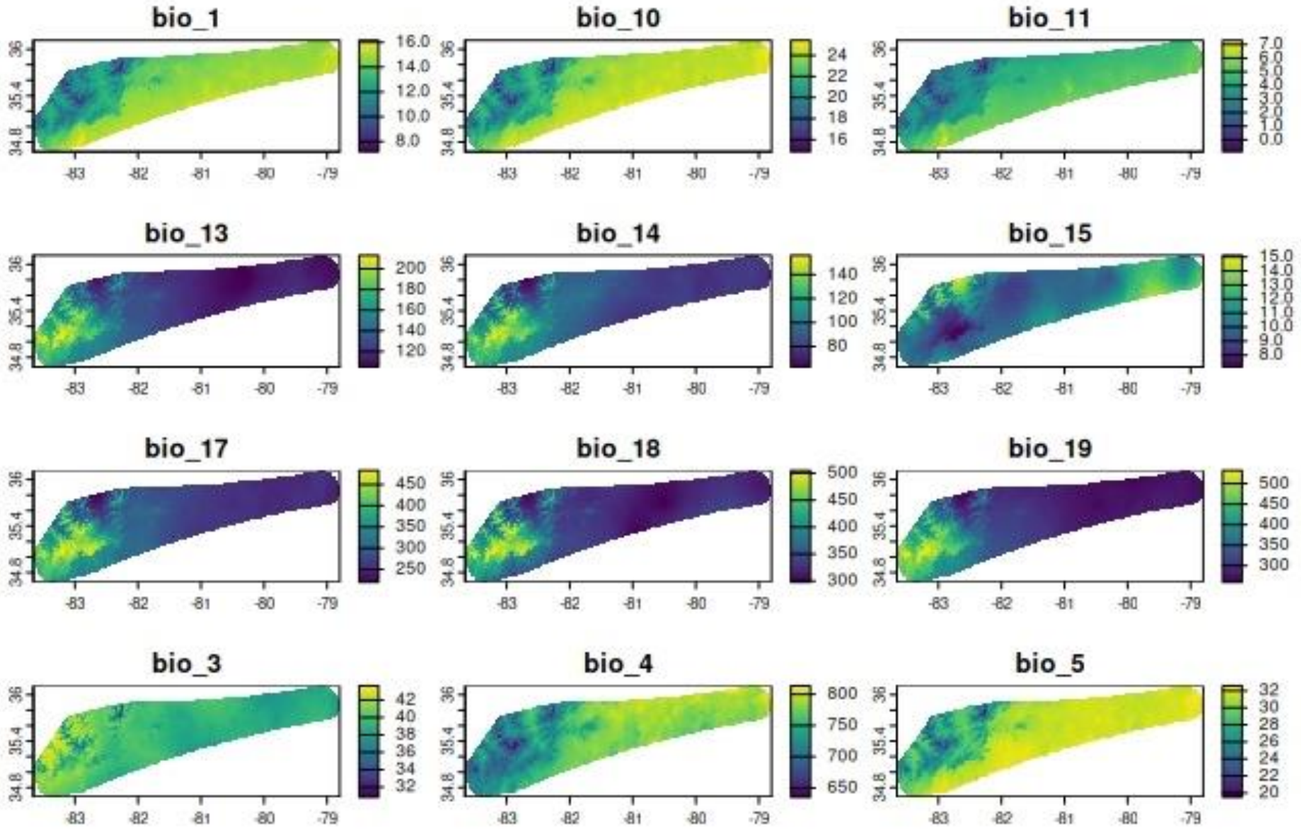
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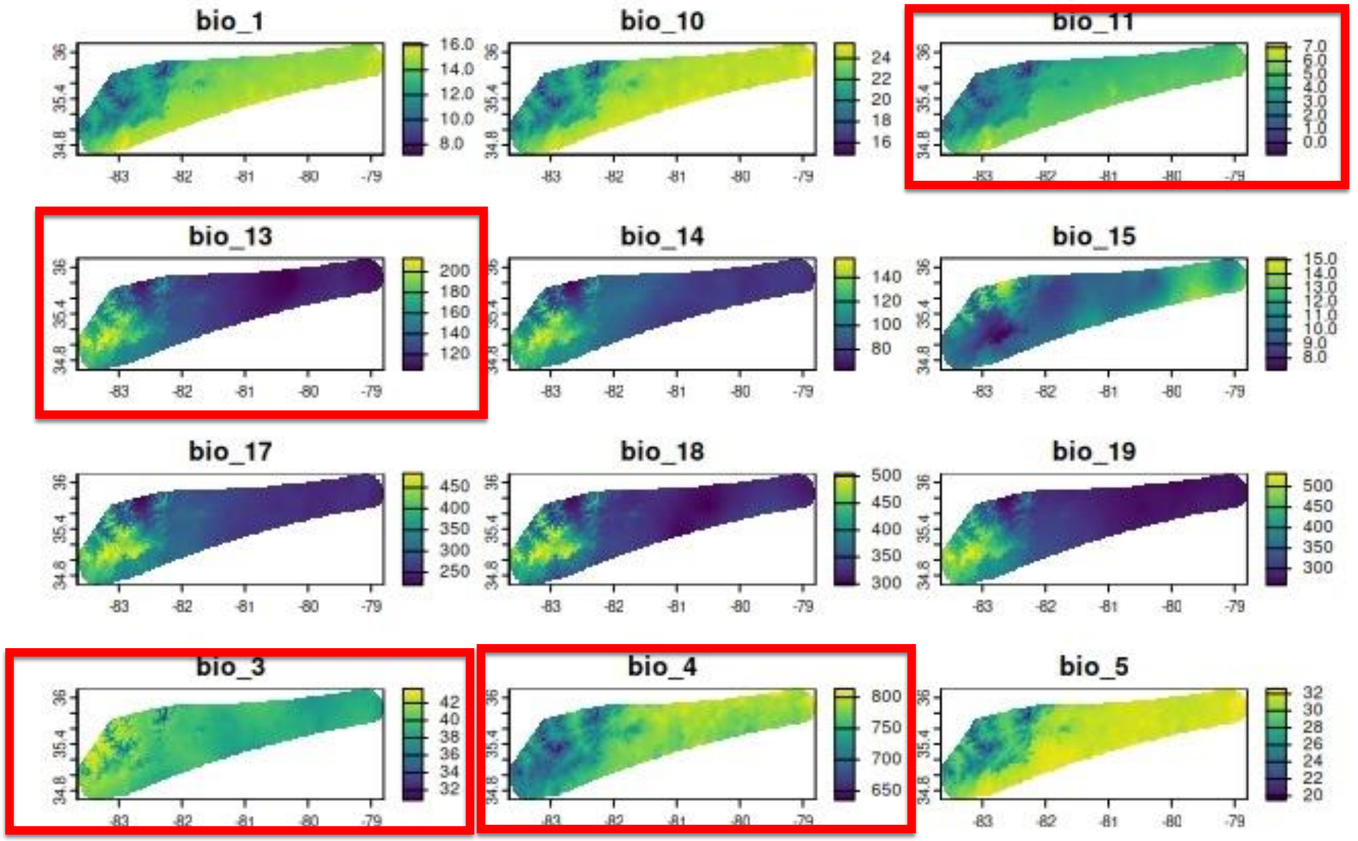
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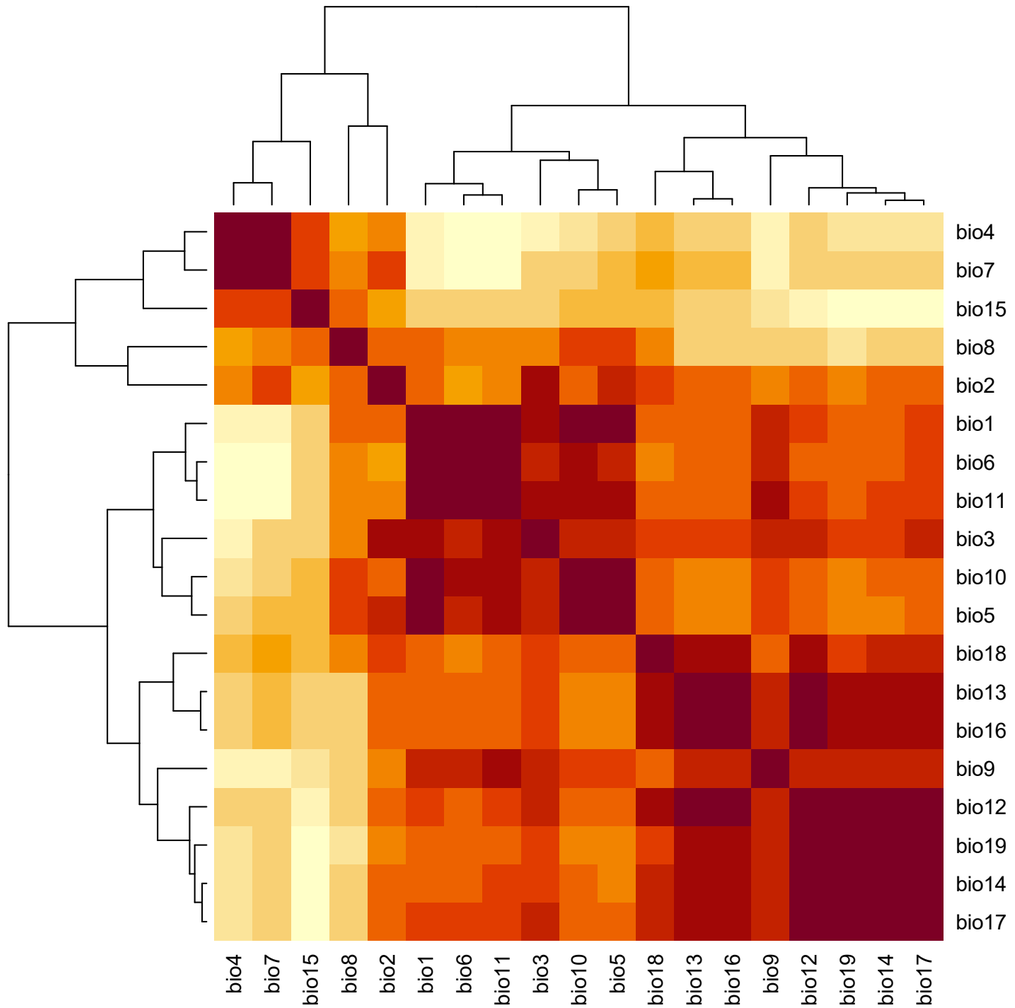
2. Obtain gridded abiotic data layers.
3. Crop layers to the accessible area.
4. **Select layers for ecological niche modeling.**



2. Select layers for ecological niche modeling

Method	Threshold	Purpose
PCA	PC1 + PC2 = ~95%	Reduces variables to axis
Pearson correlation	Removes variables > |0.80|	Removes correlated variables
Variable Inflation Factor	Removes variables > 10	Removes multicollinear variables

2. Select layers for ecological niche modeling



Layers used in ENM development

bio2
bio3
bio5
bio7
bio8
bio9
bio15
bio18
bio19

- Pearsons correlation
- A correlation test is performed on all 19 BioClim layers.
- Removes variables > |0.80|

Melton et al. 2022. Global Ecology and Biogeography.

2. Select layers for ecological niche modeling

Variance Inflation Factor (VIF)

- VIF measures **how much information is shared between predictor variables.**

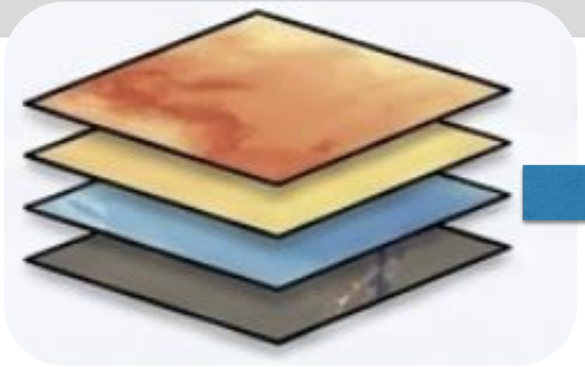
- If two or more variables are very similar, they provide redundant information.

- Detects multicollinearity.

BIO1 = Annual Mean Temperature
BIO2 = Mean Diurnal Range (Mean of monthly (max temp - min temp))
BIO3 = Isothermality (BIO2/BIO7) (×100)
BIO4 = Temperature Seasonality (standard deviation ×100)
BIO5 = Max Temperature of Warmest Month
BIO6 = Min Temperature of Coldest Month
BIO7 = Temperature Annual Range (BIO5-BIO6)
BIO8 = Mean Temperature of Wettest Quarter
BIO9 = Mean Temperature of Driest Quarter
BIO10 = Mean Temperature of Warmest Quarter
BIO11 = Mean Temperature of Coldest Quarter
BIO12 = Annual Precipitation
BIO13 = Precipitation of Wettest Month
BIO14 = Precipitation of Driest Month
BIO15 = Precipitation Seasonality (Coefficient of Variation)
BIO16 = Precipitation of Wettest Quarter
BIO17 = Precipitation of Driest Quarter
BIO18 = Precipitation of Warmest Quarter
BIO19 = Precipitation of Coldest Quarter

2. Select layers for ecological niche modeling

INPUT:RASTER STACK



STEPWISE VIF LOOP

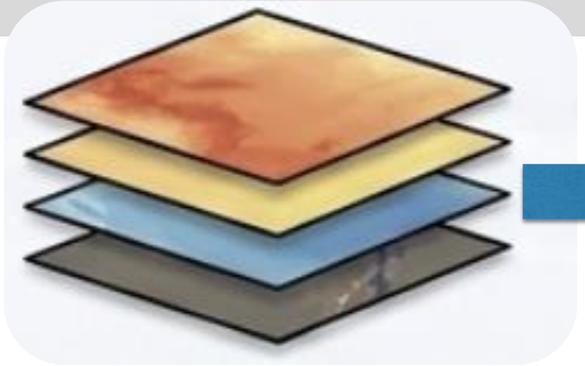
STEP 1: CALCULATE INDVIVIDUAL VIFs

STEP 2: IDENTIFY AND COMPARE HIGHEST VIFs

STEP 3: COMPARE AGANST THRESHOLD (e.g. >10)

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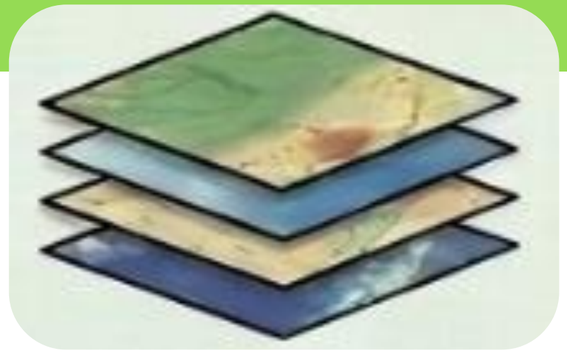
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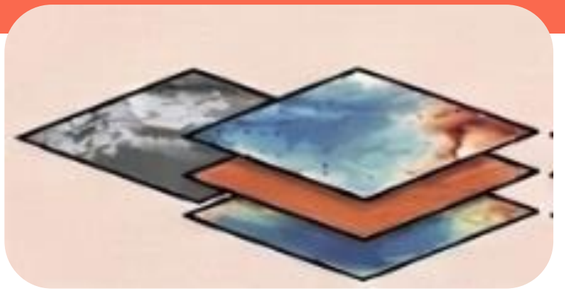
>10

<10

LOOP BACK: RE-EVALUALTE STACK

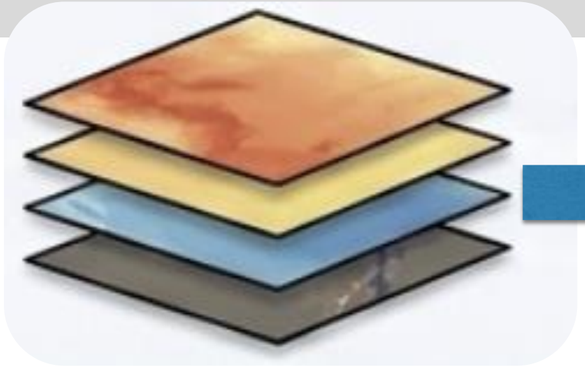


EXCLUDE REDUNDANT LAYERS



2. Select layers for ecological niche modeling

INPUT:RASTER STACK



STEPWISE VIF LOOP

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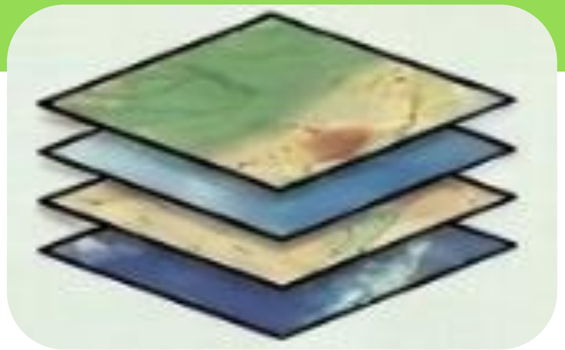
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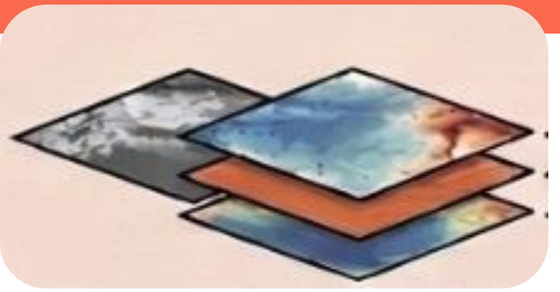
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LOOP BACK: RE-EVALUALTE STACK



EXCLUDE REDUNDANT LAYERS

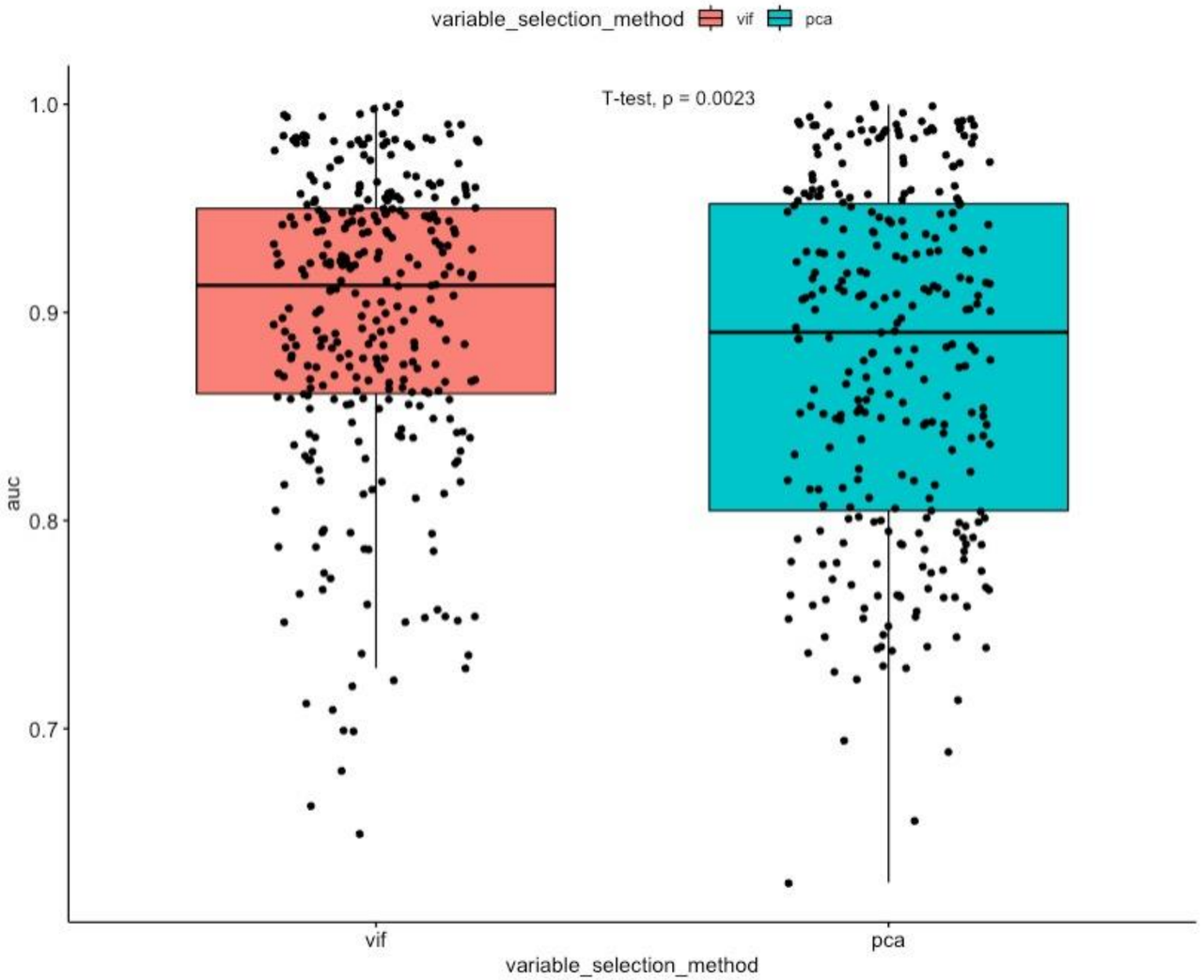


Iteration ceases when the maximum VIF value within the remaining raster stack falls below 10.

2. Select layers for ecological niche modeling

PCA vs VIF with permutation

- **AUC was significantly higher in models generated using the VIF method**
- Based on 109 species in the subfamily Crotalinae (Viperidae)
- PCA with 75% collinearity cutoff, each variable contribution had to be >5%.
- VIF < 10, permutation importance > 1%



Rautsaw et al. 2022. Scientific Data.

3. Activity

- Define the accessible area (M) for each species, crop WorldClim layers accordingly, and select environmental variables using Variance Inflation Factor (VIF) analysis.

